

2(a)(i)	(gas that obeys) $pV \propto T$ (for all values of p, V and T)	M1
	where T is thermodynamic temperature	A1
2(a)(ii)	temperature = $-273.15\text{ }^{\circ}\text{C}$	A1
2(b)(i)	$pV = NkT$	C1
	$N = (1.37 \times 10^5 \times 0.640) / (1.38 \times 10^{-23} \times (227 + 273))$	C1
	$= 1.27 \times 10^{25}$	A1
2(b)(ii)	mass = $0.0424 / (1.27 \times 10^{25})$ $= 3.34 \times 10^{-27}\text{ kg}$	A1
2(b)(iii)	$\frac{1}{2}m\langle c^2 \rangle = (3/2)kT$	C1
	$3.34 \times 10^{-27} \times v^2 = 3 \times 1.38 \times 10^{-23} \times 500$	C1
	$v = 2490\text{ m s}^{-1}$	A1
	or	
	$pV = \frac{1}{3}(Nm) \langle c^2 \rangle$ <u>and</u> Nm = mass of gas	(C1)
	$0.0424 \times v^2 = 3 \times 1.37 \times 10^5 \times 0.640$	(C1)
	$v = 2490\text{ m s}^{-1}$	(A1)
2(c)	sketch: line from (0, 0) to (500, v)	B1
	line with decreasing positive gradient throughout	B1